IntelliTrack 2.0 - Chat History Summary & System Direction

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1. Initial System Context

The capstone project is IntelliTrack 2.0, an Intelligent Submission and Compliance Tracking System for Academic Capstone Projects. The system originally included multiple workflows such as AI analysis, review pipelines, analytics, and approval systems, but later discussions narrowed the system focus toward submission compliance monitoring.

2. Core Roles Discussed

Student: - Dashboard with AI smart reminders - Submission history and tracking - Deliverable monitoring - Deadline tracking - Risk indicators Adviser: - Monitor assigned groups - View group submissions - View progress and compliance - Comment-only workflow Coordinator: - Institutional oversight dashboard - Analytics and compliance tracking - Auto-group creation during classlist import - Submission tracker and analytics - Milestone management with section-specific deadlines Admin: - Generic administrative management - User and role management

3. Major Architectural Realization

A major realization occurred during the discussion: the system should NOT function as a grading/revision/approval platform. Instead, IntelliTrack should focus on: - submission tracking - compliance monitoring - institutional oversight - deadline intelligence - analytics for late/missing submissions This caused a major architectural refactor because old workflows such as: APPROVED, REJECTED, NEEDS_REVISION, UNDER_REVIEW were deeply integrated into the backend and frontend.

4. Final Submission Workflow

Final simplified workflow: Student uploads submission ↓ System records timestamp ↓ Deadline engine evaluates compliance ↓ Status computed automatically ↓ Adviser monitors and comments ↓ Coordinator tracks institutional compliance Final statuses: - PENDING - SUBMITTED_ON_TIME - SUBMITTED_LATE - NOT_SUBMITTED - COMMENTED

5. Analytics Direction

Coordinator analytics should focus ONLY on: - on-time submissions - late submissions - missing submissions - compliance rates - adviser workload - group risk monitoring Example analytics: Group A: - 3 on-time - 1 late - 1 missing - Risk Level: HIGH

6. AI Usage Direction

AI should SUPPORT the system, not control core logic. Backend computes: - compliance metrics - late tracking - missing tracking - risk calculations AI only: - explains analytics - generates reminders - creates summaries

7. Major Refactor Recommendation

The assistant strongly recommended: - stopping isolated random fixes - performing vertical end-to-end refactors - rebuilding workflows module-by-module - aligning frontend and backend completely Suggested refactor order: 1. Student workflow 2. Adviser workflow 3. Coordinator workflow 4. Admin cleanup 5. Final alignment and deployment

8. Auto Grouping Logic

During classlist import: - detect group numbers automatically - create groups automatically - assign students automatically - update analytics automatically Coordinator classlist views should display: - student - section - adviser - assigned group

9. Milestone & Deadline Logic

Deliverables should support: - subject-specific deadlines - section-specific deadlines Example: IT411 Section A → Chapter 1 due May 25 IT411 Section B → Chapter 1 due June 1

10. AI Agent Documentation Recommendations

Multiple markdown documentation files were recommended: - AI_SYSTEM_CONTEXT.md - AI_DEVELOPMENT_RULES.md - IMPLEMENTATION_PHASES.md - VALIDATION_CHECKLIST.md - ANALYTICS_LOGIC.md These files serve as permanent context for: - Claude Sonnet - Copilot - Trae AI - VSCode AI agents

11. Security & Stability Fixes Completed

Completed fixes included: - JWT secret validation - Stateless session policy - IDOR protection - transactional upload handling - cache eviction - AI timeout handling - N+1 query optimization - logging standardization - production profile safeguards

12. Final System Identity

IntelliTrack 2.0 should now be treated as: 'An Intelligent Submission and Compliance Tracking System for Academic Capstone Projects' Main focus: - submission monitoring - compliance analytics - institutional oversight - adviser monitoring - automated grouping - AI-assisted insights